Pre-Analysis-Plan: ManyDaughters Project

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Introduction

This document outlines our pre-analysis plan (PAP) for empirically testing the four hypotheses provided by the ManyDaughters project using SOEP data. We define variables, criteria for sample selection, model specifications, and interpretation guidelines for each hypothesis.

1 Hypothesis 1: Having daughters leads to stronger support for gender equality (as measured by attitudes)

Overview

To identify the causal effect of having a daughter on support towards gender equality (as measured by attitudes), we estimate a cross-sectional OLS regression in a sample of parents who in 2018 only have exactly one child (first-born). Specifically, we use the fact that the sex of the first-born child is random by nature for the identification of the effect of having a daughter compared to having a son. Our dependent variable is surveyed in 2018 and captures attitudes towards equal sharing of employment and care work within couples. We include only individuals for whom we also have valid observations for their partner, in order to account for household fixed effects.

Variables

- **Dependent variable:** Equality attitude. Comprises attitudes toward gender equality based on `plh308_v2`. The survey question is: "It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally" Respondents are asked to respond on 7-point Likert scale from "[1] Disagree completely" to "[7] Agree completely".
- **Independent variable:** Binary indicator (*Daughter*).
- **Controls:** age, age^2, migration background, federal state, year of childbirth, education, parents' education, political party vote, labor earnings, employment status.

Variable Coding

- Outcome: *EqualityAttitude*, based on `plh308_v2` (survey year 2018), standardized to have mean 0 and SD 1.
- Daughter: binary indicator *daughter* that equals 1 if individual has a daughter and equals 0 if individual has a son, based on `kidsex` in the biobirth dataset.
- Migration background: categorial variable `migback`, differentiating between no migration background, indirect, and direct migration background.
- Federal state: categorial variable `bula`, differentiating between place of residence in one of the 16 federal states.
- Year of childbirth: numerical variable, containing the year of childbirth to control for the child's age in 2018, based on variable `kidgeb` in the biobirth dataset.
- Education: A binary indicator that equals 1 if individual has A-Levels (Abitur), based on `pgpsbil` ≥ 4.
- Parental education: binary indicator that equals 1 if one of an individual's parents has at least A-Levels (Abitur), based on `fsedu` and `msedu` in the bioparen dataset.
- Political party vote: categorial variable comprising what political party an individual voted for at the last federal election (2017) or did not vote at all, based on `plh0333`. The variable needs to be recoded for respondents from samples M1–M5 (migrants), as they were not asked the question because non-citizens are not entitled to vote in federal elections. Therefore, we recode their missing values as 'not entitled to vote'.
- monthly labor earnings: Numeric value of monthly labor earnings, including zeros for non-working individuals, based on `pglabgro` in the pgen dataset.
- Employment status: categorical variables that defines different employment statuses based on variable `pgemplst`.

Sample Criteria

- Only include individuals with exactly one child in 2018 (first born).
- Only include individuals for whom we also have valid observations for their partner and who live in the same household in 2018.
- Only include individual observations with full information for all of the variables described in the section 'Variable Coding' in 2018.

Regression Specification

$$EqualityAttitude_i = \omega + \theta_h + \beta \text{Daughter}_i + \boldsymbol{\gamma}\mathbf{X}_i + \varepsilon_i \tag{1}$$

- $EqualityAttitude_i$: Attitude toward gender equality for individual i.
- ω: constant.
- θ_h: Household fixed effects.
- Daughter_i: Indicator equal to 1 if the first child is a daughter, 0 if a son.
- $\mathbf{X}_i$: Covariates as defined in the section 'Variables'.
- ε_i: Error term.
- β: Coefficient of interest; quantifies effect of having a daughter vs having a son.
- Standard errors: Clustered on household level.

Interpretation

- Primary test statistic: coefficient β.
- If $\beta > 0$ and statistically significant ($p < 0.05$), we interpret this as support for the hypothesis. Reject otherwise.

2 Hypothesis 2: Having daughters leads to more left-leaning political preferences.

Overview

To identify the causal effect of having a daughter on political preferences, we use the panel structure of the data and employ a difference-in-differences (DiD) approach. Specifically, we compare within-individual changes in outcomes before and after the birth of a child between two groups: parents whose first child is a daughter (treatment group) and those whose first child is a son (control group). As the gender of the first-born child is random by nature, we can exploit this exogenous variation to identify causal treatment effects. We use a Two-Way-Fixed-Effect setting accounting for individual, household, and survey year fixed effects. To measure political preference, we use a variable in which individuals indicate their political views on a Likert scale ranging from 0 (far left) to 10 (far right).

Variables

- **Dependent variable:** *Political attitudes* (SOEP: `plh0004`[1]). Stated on Likert Scale where [0] = far left and [10] = far right attitudes.
- **Independent variable:** Binary indicator for having a daughter (*Daughter*), binary indicator for periods after treatment (*Post*), interaction term $Daughter * Post$.
- **Controls:** age, age^2, education, labor earnings, employment status.

Variable Coding

- Political attitudes: Values from initial Likert scale are standardized to have mean = 0 and SD = 1.
- Daughter: binary indicator *Daughter* that equals 1 if individual has a daughter and equals 0 if individual has a son, based on `kidsex` in the biobirth dataset.
- Post: A binary indicator that equals 1 if survey year $\geq$ year of birth (first child).
- Education: A binary indicator that equals 1 if individual has A-Levels (Abitur), based on `pgpsbil` ≥ 4.
- Monthly labor earnings: Numeric value of monthly labor earnings, including zeros for non-working individuals, based on `pglabgro` in the pgen dataset.
- Employment status: categorical variables that defines different employment statuses based on variable `pgemplst`.

Sample Criteria

- Only include individuals with no change in family status e.g. no separation/divorce.
- Only include individual observations with full information for all of the variables described in the section 'Variable Coding' per survey year.
- Restrict to years: Our dependent variable is surveyed in 2005, 2009, 2014, and 2019. We thus consider the period 2005-2019 as our possible treatment period.
- Only include individuals with at least one valid pre- and post treatment observation.
- Only include individuals with the first biological child. Omit data on individuals as soon as a second child is born.

[1] Question asks: *'In politics people often talk about "left" and "right" when it comes to characterize different political attitudes. If you think about your own political views: Where would you place yours?'.*

Regression Specification

$$PoliticalAttitude_{it} = \omega + \alpha_i + \theta_h + \lambda_t + \beta_1 \text{Post}_{it} + \beta_2 \text{Daughter}_i + \beta_3(\text{Post}_{it} \times \text{Daughter}_i) + \boldsymbol{\gamma}\mathbf{X}_{it} + \varepsilon_{it} \tag{2}$$

- $PoliticalAttitude_{it}$: Standardized political attitude for individual i at time t.
- ω: constant.
- α_i: Individual fixed effects.
- θ_h: Household fixed effects.
- λ_t: Year fixed effects.
- Post_{it}: Indicator equal to 1 in periods of and after the birth of the first child, 0 otherwise..
- Daughter_i: Indicator equal to 1 if the first child is a daughter, 0 if a son.
- $\text{Post}_{it} \times \text{Daughter}_i$: Interaction term identifying post treatment effect.
- $\mathbf{X}_{it}$: Covariates as defined in the section 'Variables'.
- ε_{it}: Error term.
- β_3: Coefficient of interest; quantifies post treatment effect.
- Standard errors: Clustered on household level

Interpretation

- Primary test statistic: coefficient β_3
- If $\beta_3 < 0$ and statistically significant ($p < 0.05$), we interpret this as support for the hypothesis. Reject otherwise.

3 Hypothesis 3: Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes)

Overview

To identify the causal effect of having a daughter on support towards gender equality in the household (as measured by behaviours), we use the panel structure of the data and employ a difference-in-differences (DiD) approach. Specifically, we compare within-individual changes in support towards gender equality before and after the birth of a child between two groups: parents whose first child is a daughter (treatment group) and those whose first child is a

son (control group). As the gender of the first-born child is random by nature, we can exploit this exogenous variation to identify causal treatment effects. We use a Two-Way-Fixed-Effect setting accounting for individual, household, and survey year fixed effects. To measure support for gender equality (by behaviors), we use a variable that states the average weekday amount of time an individual spends on all of the following tasks compared to their partner: running errands, household chores, child care and care for other persons.

Variables

- **Dependent variable:** Time Allocation. States the average weekday amount of time an individual spends on all of the following tasks compared to their partner: running errands (SOEP:`pli0040`[2]), household chores (SOEP:`pli0043_h`), child care (SOEP:`pli0044_h`), care for persons (SOEP:`pli0046`).
- **Independent variable:** Binary indicator for having a daughter ($Daughter$), binary indicator for periods after treatment ($Post$), interaction term $Daughter * Post$.
- **Controls:** age, age^2, education, labor earnings, employment status.

Variable Coding

- Time Allocation: Relative amounts are calculated as:

$$TimeAllocation_{it} = \frac{T_{it}}{T_{it} + T_{pt}} \tag{3}$$

  - Where T_{it} denotes the average daily amount of time (in hours) individual i spends on the stated tasks in period t.
  - Where T_{pt} denotes the average daily amount of time (in hours) individual i's partner p spends on the stated tasks in period t.

- Daughter: binary indicator `Daughter` that equals 1 if individual has a daughter and equals 0 if individual has a son, based on `kidsex` in the biobirth dataset.
- Post: A binary indicator that equals 1 if survey year $\geq$ year of birth (first child).
- Education: A binary indicator that equals 1 if individual has A-Levels (Abitur), based on `pgpsbil` ≥ 4.
- Monthly labor earnings: Numeric value of monthly labor earnings, including zeros for non-working individuals, based on `pglabgro` in the pgen dataset.
- Employment status: categorical variables that defines different employment statuses based on variable `pgemplst`.

[2]Question asks: *What is a typical day like for you? How many hours do you spend on the following activities on a typical weekday?* Activities include: running errands, household chores, child care, care for persons.

Sample Criteria

- Only include individuals with the first biological child. Omit data on individuals as soon as a second child is born.
- Only include individuals with no change in family status e.g. no separation/divorce.
- Only include individual observations with full information for all of the variables described in the section 'Variable Coding' per survey year.
- Only include individuals where T_{it} and T_{pt} is defined (full informations on individual and partners time allocation).
- Only include individuals with at least one valid pre- and post treatment observation.
- Restrict to years: Although the time allocation variables are collected annually, we restrict the sample to the years 2005–2019 to ensure a comparable time period with H1, H2, and H4.

Regression Specification

$$TimeAllocation_{it} = \omega + \alpha_i + \theta_h + \lambda_t + \beta_3 \text{Post}_{it} + \beta_2 \text{Daughter}_i + \beta_3 (\text{Post}_{it} \times \text{Daughter}_i) + \boldsymbol{\gamma} \mathbf{X}_{it} + \varepsilon_{it} \tag{4}$$

- $TimeAllocation_{it}$: Relative amount of time individual i spends on sated tasks compared to their partner p in period t.
- ω: constant.
- α_i: Individual fixed effects.
- θ_h: Household fixed effects.
- λ_t: Year fixed effects.
- Post_{it}: Indicator equal to 1 in periods of and after the birth of the first child, 0 otherwise.
- Daughter_i: Indicator equal to 1 if the first child is a daughter, 0 if a son.
- $\text{Post}_{it} \times \text{Daughter}_i$: Interaction term identifying post treatment effect.
- $\mathbf{X}_{it}$: Covariates as defined in the section 'Variables'.
- ε_{it}: Error term.
- β_3: Coefficient of interest; quantifies post treatment effect.
- Standard errors: Clustered on household level.

Interpretation

- Primary test statistic: coefficient β_3
- If $\beta_3 > 0$ and statistically significant ($p < 0.05$), we interpret this as support for the hypothesis. Reject otherwise.

4 Hypothesis 4: Having daughters leads to a larger increase in the support of gender equality for men than for women

Overview

To identify the causal effect of having a daughter on support towards gender equality (as measured by attitudes), we estimate a cross-sectional OLS regression in a sample of parents who in 2018 only have exactly one child (first-born). We estimate a similar specification to Hypothesis 1, but fully interacted with a sex dummy, differentiating mothers from fathers. Specifically, we use the fact that the sex of the first-born child is random by nature for the identification of the effect of having a daughter compared to having a son. Our dependent variable is surveyed in 2018 and captures attitudes towards equal sharing of employment and care work within couples. The coefficient of interest captures the interaction term between father/mother and daughter/son. We include only individuals for whom we also have valid observations for their partner, in order to account for household fixed effects.

Variables

- **Dependent variable:** Equality attitude. Comprises attitudes toward gender equality based on `plh308_v2`. The survey question is: "It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally" Respondents are asked to respond on 7-point Likert scale from "[1] Disagree completely" to "[7] Agree completely".
- **Independent variable:** Binary indicator ($Daughter$), binary indicator ($Male$), interaction term $Daughter \times Male$.
- **Controls:** age, age^2, migration background, federal state, year of childbirth, education, parents' education, political party vote, labor earnings, employment status. All of the stated control variables interacted with $Male$.

Variable Coding

- Outcome: `EqualityAttitude`, based on `plh308_v2` (survey year 2018), standardized to have mean = 0 and SD = 1.
- Daughter: binary indicator `daughter` that equals 1 if individual has a daughter and equals 0 if individual has a son, based on `kidsex` in the biobirth dataset.

- Migration background: categorial variable `migback`, differentiating between no migration background, indirect, and direct migration background.
- Federal state: categorial variable `bula`, differentiating between place of residence in one of the 16 federal states.
- Year of childbirth: numerical variable, containing the year of childbirth to control for the child's age in 2018, based on variable `kidgeb` in the biobirth dataset.
- Education: A binary indicator that equals 1 if individual has A-Levels (Abitur), based on `pgpsbil` ≥ 4.
- Parental education: binary indicator that equals 1 if one of an individual's parents has at least A-Levels (Abitur), based on `fsedu` and `msedu` in the bioparen dataset.
- Political party vote: categorial variable comprising what political party an individual voted for at the last federal election or did not vote at all, based on `plh0333`. Variable has to be recoded for respondents from sample M1-M5. Migrant samples were not asked because non-citizens are not entitled to vote in federal elections.
- Monthly labor earnings: Numeric value of monthly labor earnings, including zeros for non-working individuals, based on `pglabgro` in the pgen dataset.
- Employment status: categorical variables that defines different employment statuses based on variable `pgemplst`.

Sample Criteria

- Only include individuals with exactly one child in 2018 (first born).
- Only include individuals for whom we also have valid observations for their partner and who live in the same household in 2018.
- Only include individual observations with full information for all of the variables described in the section 'Variable Coding' in 2018.

Regression Specification

$$\begin{aligned} &EqualityAttitude_i = \omega + \theta_h + \beta_1 \text{Daughter}_i + \beta_2 \text{Male} \\ &+\beta_3(\text{Daughter}_i \times \text{Male}_i) + \boldsymbol{\gamma}\mathbf{X}_i + \boldsymbol{\eta}(\mathbf{X}_i \times \text{Male}_i) + \varepsilon_i \end{aligned} \tag{5}$$

- $EqualityAttitude_i$: Attitude toward gender equality for individual i.
- ω: constant.
- θ_h: Household fixed effects.
- Daughter_i: Indicator equal to 1 if the first child is a daughter, 0 if a son.

- $\mathbf{X}_i$: Covariates as defined in the section 'Variables'.
- ε_i: Error term.
- β_1: quantifies the overall effect of having a daughter vs having a son.
- β_2: quantifies differences in attitudes towards gender equality between fathers and mothers.
- β_3: Coefficient of interest; captures the difference between the effect of having a daughter instead of a son on fathers compared to mothers.
- Standard errors: Clustered on household level.

Interpretation

- Primary test statistic: coefficient β_3.
- If $\beta_3 > 0$ and statistically significant ($p < 0.05$), we interpret this as support for the hypothesis. Reject otherwise.